

#1 · Protect Self and Others

What does WHMIS stand for?

Answer

Workplace Hazardous Materials Information System. It is the Canada wide system for labeling hazardous products and providing Safety Data Sheets so workers know what they are handling.

#2 · Protect Self and Others

What is a Safety Data Sheet (SDS)?

Answer

A detailed document for a hazardous product that covers ingredients, first aid steps, fire fighting measures, and safe handling and storage instructions.

#3 · Protect Self and Others

What is a workplace label?

Answer

A label added by shop staff to a secondary container, such as a bottle filled from a larger drum, showing the product name and safety information.

#4 · Protect Self and Others

What does a WHMIS pictogram communicate?

Answer

It shows the general type of hazard, such as flammability or corrosion, at a glance through a picture inside a red diamond border.

#5 · Protect Self and Others

Which WHMIS pictogram usually appears on flammable aerosol primers and thinners?

Answer

The flame symbol, warning that the product can ignite easily from a spark or open flame.

#6 · Protect Self and Others

Where must the shop's SDS binder or system be kept?

Answer

Somewhere every worker can access it at any time during their shift, not locked away or kept offsite.

#7 · Protect Self and Others

Why is PPE considered the last line of defense?

Answer

Because a shop should try to remove or control a hazard at its source first, with PPE protecting the worker from whatever risk is left over.

#8 · Protect Self and Others

What eye protection is needed for grinding and cutting?

Answer

A full face shield worn together with safety glasses, since grinding throws fragments across a wide area and can strike the whole face.

#9 · Protect Self and Others

Why are leather gloves preferred for metal cutting and grinding?

Answer

Leather resists sharp edges, sparks, and heat better than thin nitrile gloves, which are meant for chemical handling instead.

#10 · Protect Self and Others

What glove type is correct for mixing chemical adhesives or paint products?

Answer

Nitrile gloves, since they are rated for chemical resistance and protect the skin from liquid chemical exposure.

#11 · Protect Self and Others

Why must loose clothing and jewelry be avoided around spinning tools?

Answer

They can catch on moving parts like a buffer wheel or drill and pull the worker's hand or body into the tool, causing serious injury.

#12 · Protect Self and Others

Why is hearing protection required during long grinding and sanding tasks?

Answer

Repeated loud noise causes permanent hearing damage that builds up gradually over a career, even on days that do not feel unbearably loud.

#13 · Protect Self and Others

Why is a paper dust mask not enough for spray painting?

Answer

It only filters larger dust particles and cannot filter chemical vapors released during solvent based spray painting.

#14 · Protect Self and Others

What is the minimum respiratory protection for solvent based painting?

Answer

A half face air purifying respirator fitted with organic vapor cartridges.

#15 · Protect Self and Others

Why are isocyanate containing clear coats especially dangerous?

Answer

They can sensitize the lungs over repeated exposure, so even a tiny future exposure can trigger a severe allergic reaction.

#16 · Protect Self and Others

What respirator system is used for isocyanate based spraying?

Answer

A supplied air respirator system that pumps clean outside air to the painter, instead of relying on filter cartridges alone.

#17 · Protect Self and Others

What does it mean if a painter smells solvent through their respirator?

Answer

The filter cartridge is worn out and needs to be replaced immediately.

#18 · Protect Self and Others

Why does respirator fit testing matter?

Answer

It confirms the respirator seals properly against an individual worker's face, since face shape and facial hair affect the seal.

#19 · Protect Self and Others

What is the main safety purpose of spray booth exhaust ventilation?

Answer

It keeps flammable vapor concentration low enough to avoid ignition and reduces harmful vapor exposure to the painter.

#20 · Protect Self and Others

Why are grounding straps and bonding used when transferring flammable liquids?

Answer

They safely discharge static electricity that could otherwise spark and ignite flammable vapors.

#21 · Protect Self and Others

Why is smoking or an open flame near a spray booth dangerous, even outside the door?

Answer

Flammable vapors can drift beyond the booth opening and still reach and ignite near an open flame nearby.

#22 · Protect Self and Others

What happens when spray booth filters are clogged and not replaced?

Answer

Airflow is reduced and combustible overspray material builds up in the ductwork, increasing fire risk.

#23 · Protect Self and Others

Why must spray booth lighting be sealed or explosion proof rated?

Answer

A broken bulb or electrical fault could otherwise become an ignition source inside a vapor filled booth.

#24 · Protect Self and Others

What does a spray booth interlock system do?

Answer

It prevents the spray gun or heater from operating unless the exhaust fan is confirmed running.

#25 · Protect Self and Others

Why must solvent soaked rags go in a closed metal container?

Answer

They can heat up through spontaneous combustion and ignite on their own without any external flame nearby.

#26 · Protect Self and Others

Where should used oil and other liquid hazardous waste be stored?

Answer

In labeled, sealed containers kept away from floor drains, to prevent spills from reaching the environment.

#27 · Protect Self and Others

What color are flammable liquid safety storage cabinets typically painted?

Answer

Yellow, so they can be quickly and clearly identified in the shop.

#28 · Protect Self and Others

Are empty aerosol cans and spent spray filters hazardous waste?

Answer

Yes, until confirmed empty and handled by a licensed waste hauler, since residual product can remain inside.

#29 · Protect Self and Others

Why does a shop need records of hazardous waste pickup and disposal?

Answer

Ontario environmental regulations require a documented paper trail, often called a manifest, showing where the waste went.

#30 · Protect Self and Others

Why should hazardous waste containers be kept away from floor drains?

Answer

A spill near a drain can quickly contaminate groundwater by reaching an outside sewer or storm system.

#31 · Protect Self and Others

Why is sanding dust from primer and filler a fire hazard?

Answer

It is combustible and can accumulate as fuel on ledges, fixtures, and inside equipment over time.

#32 · Protect Self and Others

Which fire extinguisher class is used for paint and solvent fires?

Answer

Class B, which is rated for flammable liquid fires.

#33 · Protect Self and Others

Why must emergency exits stay clear of parts and clutter?

Answer

So workers can evacuate quickly and safely if something goes wrong, even for a short blockage.

#34 · Protect Self and Others

What is required before starting hot work like welding near other materials?

Answer

A posted fire watch and clearing of nearby flammable materials before work begins.

#35 · Protect Self and Others

Why can welding sparks be dangerous even a short distance away?

Answer

Sparks can travel farther than expected and ignite combustible material unnoticed, smoldering before becoming a visible fire.

#36 · Protect Self and Others

How should oxygen and acetylene cylinders be stored?

Answer

Upright, chained to prevent tipping, and kept separated from each other when not in use.

#37 · Protect Self and Others

What is the purpose of lockout and tagout (LOTO)?

Answer

To prevent equipment from starting up unexpectedly while a worker is servicing or adjusting it.

#38 · Protect Self and Others

Who should normally remove a lock placed during a lockout procedure?

Answer

The worker who applied the lock, once their task is complete and the area is confirmed clear.

#39 · Protect Self and Others

What risk does removing a machine guard create?

Answer

An increased risk of a serious hand injury from contact with the moving part the guard was blocking.

#40 · Protect Self and Others

Why is LOTO especially important for frame machines and compressors?

Answer

An unexpected startup during maintenance on this powerful equipment could crush or entangle a worker.

#41 · Protect Self and Others

What must be confirmed before repairing shop equipment itself?

Answer

That the equipment is de-energized, tagged, and confirmed clear by other workers in the shop.

#42 · Protect Self and Others

What does a machine guard on a bench grinder do?

Answer

It physically keeps hands and loose clothing away from the spinning grinding wheel.

#43 · Protect Self and Others

Why must vehicle lift points match the manufacturer's recommended locations?

Answer

The wrong lift points can cause the vehicle to slip, tip, or suffer frame or suspension damage while raised.

#44 · Protect Self and Others

What should always be engaged before working underneath a raised vehicle?

Answer

The lift's mechanical safety locks, since hydraulic pressure alone can fail silently over time.

#45 · Protect Self and Others

Why should jack stands never sit on wood blocks or uneven concrete?

Answer

An unstable base can cause the jack stand to shift or fail under the vehicle's load.

#46 · Protect Self and Others

What should be checked on a ladder before each use?

Answer

Cracks, loose rungs, or damaged feet.

#47 · Protect Self and Others

Why does a ladder's duty rating matter?

Answer

It reflects the maximum weight the ladder can safely support, including the worker plus any tools carried.

#48 · Protect Self and Others

What is the correct technique for manually lifting a heavy panel?

Answer

Bend at the knees and keep the load close to the body, rather than bending at the waist.

#49 · Protect Self and Others

What is the internal responsibility system under the OHSA?

Answer

Everyone in the workplace, from apprentice to owner, shares responsibility for safety, not just one designated person.

#50 · Protect Self and Others

What are the three basic worker rights under the OHSA?

Answer

The right to know about hazards, the right to participate in safety decisions, and the right to refuse unsafe work.

#51 · Protect Self and Others

What is the correct first step to refuse unsafe work?

Answer

Notify a supervisor immediately and cooperate with the investigation process, rather than simply walking off the job.

#52 · Protect Self and Others

What does a joint health and safety committee do?

Answer

It inspects the workplace and brings forward safety concerns, made up of both worker and management representatives.

#53 · Protect Self and Others

What should happen to an accident scene after a critical injury?

Answer

It should not be disturbed until the investigation allows it, except to help the injured person or prevent further harm.

#54 · Protect Self and Others

Who must a critical injury in an Ontario shop be reported to?

Answer

The Ministry of Labour.

#55 · Protect Self and Others

What is the definition of a hazard?

Answer

Anything in the workplace that could cause harm to a person, whether physical, chemical, or ergonomic.

#56 · Protect Self and Others

Give an example of an ergonomic hazard in an auto body shop.

Answer

Repeatedly bending over a fender for hours during bodywork, which strains the body gradually over time.

#57 · Protect Self and Others

Why do workers often ignore early signs of ergonomic injury?

Answer

Because ergonomic injuries build up slowly over months instead of happening all at once like a sudden injury.

#58 · Protect Self and Others

Why is compressed air considered a physical hazard?

Answer

High pressure air directed at skin or used to clean clothing can force air into the body and cause serious injury.

#59 · Protect Self and Others

What are the three main hazard categories covered in this chapter?

Answer

Physical hazards, chemical hazards, and ergonomic hazards, each requiring different prevention and control methods.

#60 · Protect Self and Others

What electrical hazards exist around welders and battery chargers in a shop?

Answer

Wet hands, frayed cords, or equipment used near flammable vapors can all lead to shock or fire.

#61 · Detail Exterior and Interior Vehicle Body and Components

Why is a pre-repair walkaround inspection performed before any repair work begins?

Answer

To spot and document all visible damage, including anything missed on the insurance estimate, before the vehicle is moved into the work bay.

#62 · Detail Exterior and Interior Vehicle Body and Components

What can uneven panel gaps between the hood and fenders indicate during a pre-repair inspection?

Answer

Possible hidden frame damage or a panel that was misaligned during a previous repair.

#63 · Detail Exterior and Interior Vehicle Body and Components

Why does the shop photograph and document damage before starting a repair?

Answer

It protects both the shop and the customer by proving the vehicle's condition before any work began, preventing later disputes.

#64 · Detail Exterior and Interior Vehicle Body and Components

What is a swirl mark?

Answer

A fine circular scratch that appears as a hazy pattern under bright light, usually caused by wiping dirt across paint with a dry or dirty cloth.

#65 · Detail Exterior and Interior Vehicle Body and Components

What is the purpose of the pre-rinse stage of an exterior wash?

Answer

Plain water knocks loose dirt and grit off the surface before any cloth or sponge touches the paint, preventing scratches.

#66 · Detail Exterior and Interior Vehicle Body and Components

What makes a pH-balanced car wash soap different from household dish soap?

Answer

It cleans the surface without stripping away wax or sealant protection, unlike a harsh dish soap.

#67 · Detail Exterior and Interior Vehicle Body and Components

How does the two-bucket wash method reduce paint scratching?

Answer

The wash mitt is rinsed in a plain water bucket before returning to the soap bucket, so grit is not carried back onto the paint.

#68 · Detail Exterior and Interior Vehicle Body and Components

Why should a vehicle be dried promptly after rinsing rather than left to air dry?

Answer

Air drying allows minerals in the water to evaporate on the paint and leave visible water spots.

#69 · Detail Exterior and Interior Vehicle Body and Components

What are bonded contaminants?

Answer

Particles like tree sap, tar, and rail dust that stick to paint and do not wash away with soap and water.

#70 · Detail Exterior and Interior Vehicle Body and Components

What tool is used to remove bonded contaminants from paint before polishing?

Answer

A clay bar, used with a lubricant and light pressure to lift contaminants off the surface.

#71 · Detail Exterior and Interior Vehicle Body and Components

What should be done with a clay bar that has been dropped on the shop floor?

Answer

It should be discarded, since embedded grit could scratch the paint on the next pass.

#72 · Detail Exterior and Interior Vehicle Body and Components

Why is claying normally done before polishing a vehicle?

Answer

Leaving bonded contaminants on the surface would cause the polishing pad to drag grit across the paint.

#73 · Detail Exterior and Interior Vehicle Body and Components

What is the difference between compound and polish?

Answer

Compound is a heavier abrasive used first on deep scratches and oxidation; polish is finer and used after compound or for light defects.

#74 · Detail Exterior and Interior Vehicle Body and Components

Why is a dual action polisher generally safer than a single action rotary polisher?

Answer

Its spinning and oscillating motion together reduces the risk of generating too much heat and burning through the clear coat.

#75 · Detail Exterior and Interior Vehicle Body and Components

What does a panel getting hot to the touch during polishing usually mean?

Answer

The clear coat is being cut too aggressively and risks burning through if the repairer keeps working the same spot.

#76 · Detail Exterior and Interior Vehicle Body and Components

What is oxidation on automotive paint?

Answer

A dull, chalky, faded appearance caused by long-term sun and weather exposure without regular waxing.

#77 · Detail Exterior and Interior Vehicle Body and Components

What is a fish eye defect and what usually causes it?

Answer

A small round crater in fresh paint, usually caused by silicone or oil contamination on the surface before painting.

#78 · Detail Exterior and Interior Vehicle Body and Components

What is orange peel and what usually causes it?

Answer

A bumpy, textured paint surface resembling an orange skin, usually caused by spray gun settings or paint booth conditions.

#79 · Detail Exterior and Interior Vehicle Body and Components

What ingredient in traditional wax is known for producing a deep, warm shine?

Answer

Carnauba, a natural wax derived from the leaves of a type of palm tree.

#80 · Detail Exterior and Interior Vehicle Body and Components

How does synthetic paint sealant generally compare to traditional carnauba wax?

Answer

Sealant typically lasts longer under sun and weather exposure, though it can look slightly less warm than wax.

#81 · Detail Exterior and Interior Vehicle Body and Components

What environment does ceramic coating need for a proper application?

Answer

A controlled, dust-free space with adequate time allowed to cure undisturbed before the vehicle gets wet.

#82 · Detail Exterior and Interior Vehicle Body and Components

Why should leather upholstery be cleaned with a dedicated leather cleaner and conditioner?

Answer

Harsh chemicals or excess water can dry out and crack leather, unlike more resilient cloth upholstery.

#83 · Detail Exterior and Interior Vehicle Body and Components

What method removes a deep, set-in stain from cloth upholstery?

Answer

Extracting the stain with fabric cleaner and a wet vacuum machine that pulls out moisture and dirt together.

#84 · Detail Exterior and Interior Vehicle Body and Components

Why is a matte or satin dashboard protectant generally preferred over a heavy glossy one?

Answer

A very glossy dashboard finish can create distracting glare on the windshield while driving.

#85 · Detail Exterior and Interior Vehicle Body and Components

Why does interior glass need a dedicated glass cleaner instead of a standard shop rag?

Answer

A film from off-gassing plastics can smear under a regular cloth, so a proper glass cleaner and lint-free cloth are needed.

#86 · Detail Exterior and Interior Vehicle Body and Components

What steps address a strong smoke odor in a vehicle's interior?

Answer

Ozone treatment combined with thorough cleaning of the actual source of the odor, not just masking it with air freshener.

#87 · Detail Exterior and Interior Vehicle Body and Components

What should be protected before spraying degreaser in an engine bay?

Answer

Sensitive electrical components such as the battery terminals, alternator, and open air intakes should be covered or bagged.

#88 · Detail Exterior and Interior Vehicle Body and Components

Why should high-pressure water never be aimed directly at a wiring harness during engine bay rinsing?

Answer

Forced water intrusion can cause corrosion or electrical faults in the connectors over time.

#89 · Detail Exterior and Interior Vehicle Body and Components

What components should engine dressing product never be applied to?

Answer

Belts, pulleys, or anything that could become slippery and create a mechanical hazard.

#90 · Detail Exterior and Interior Vehicle Body and Components

What lighting condition should be used for a proper final paint inspection?

Answer

Strong, even natural daylight or a dedicated inspection light, since dim fluorescent lighting can hide defects.

#91 · Prepare Vehicle for Refinishing

What is raking light used for during a pre-refinishing inspection?

Answer

Light shone at a low angle across a panel to reveal small dents, waves, and sand scratches that overhead lighting hides.

#92 · Prepare Vehicle for Refinishing

What is the difference between surface rust and perforated rust?

Answer

Surface rust sits on top of bare metal and can be sanded away; perforated rust has eaten through the metal and needs panel replacement or repair.

#93 · Prepare Vehicle for Refinishing

What does delamination mean on a painted panel?

Answer

Paint separating cleanly from the layer of material underneath it, meaning the failing coating must be removed before refinishing.

#94 · Prepare Vehicle for Refinishing

What does chalking look like and what causes it?

Answer

A dull, powdery surface caused by ultraviolet sun damage that signals the old finish has broken down.

#95 · Prepare Vehicle for Refinishing

What is galvanized steel and why does it matter for preparation?

Answer

Steel coated in zinc to resist rust; it needs a different primer approach than plain cold rolled steel.

#96 · Prepare Vehicle for Refinishing

What happens to sandpaper scratches as grit number increases?

Answer

The abrasive particles get finer and the scratches left behind get shallower.

#97 · Prepare Vehicle for Refinishing

What problem happens if a technician jumps from 80 grit straight to 400 grit?

Answer

The coarse 80 grit scratches remain because the finer paper cannot reach deep enough to remove them.

#98 · Prepare Vehicle for Refinishing

What is featheredging?

Answer

Sanding the edge of a repair in a wide, gradual taper so it blends smoothly into surrounding paint with no step.

#99 · Prepare Vehicle for Refinishing

How can a technician tell if a featheredge is done correctly?

Answer

A good featheredge cannot be felt with a fingernail run across it.

#100 · Prepare Vehicle for Refinishing

What happens if an unfeathered hard step is painted over anyway?

Answer

The step telegraphs, showing through the new paint as a visible ridge or line.

#101 · Prepare Vehicle for Refinishing

Why are DA (dual action) sanders preferred over single direction sanders?

Answer

Their random orbital motion reduces the chance of visible swirl marks in the finish.

#102 · Prepare Vehicle for Refinishing

What is tooth or profile in surface preparation?

Answer

The roughened texture sanding creates so primer and paint have something to mechanically grip onto.

#103 · Prepare Vehicle for Refinishing

What grit range is typical right before applying primer over shaped filler?

Answer

Around 320 to 360 grit.

#104 · Prepare Vehicle for Refinishing

What grit range is typical for sanding cured primer before color?

Answer

Around 400 to 600 grit.

#105 · Prepare Vehicle for Refinishing

What is the main purpose of masking before refinishing work?

Answer

To protect glass, trim, and adjacent panels from overspray.

#106 · Prepare Vehicle for Refinishing

What happens if masking tape is left with air bubbles or loose edges?

Answer

Paint can bleed underneath it, leaving a ragged, unwanted paint line.

#107 · Prepare Vehicle for Refinishing

What is reverse masking, or hinging?

Answer

Folding a lip of tape back on itself along a curved edge so paper hangs away from the surface.

#108 · Prepare Vehicle for Refinishing

Why should sensors on a bumper cover be masked or removed before spraying?

Answer

Paint on a sensor can interfere with how it works.

#109 · Prepare Vehicle for Refinishing

What is the main purpose of wax and grease remover?

Answer

A solvent cleaner that dissolves and lifts oils, wax, silicone, and tar that sanding alone cannot remove.

#110 · Prepare Vehicle for Refinishing

Why is a second clean, dry cloth used to wipe off wax and grease remover?

Answer

To prevent dissolved contamination from redepositing back onto the panel before the solvent flashes off.

#111 · Prepare Vehicle for Refinishing

What paint defect can result from letting wax and grease remover air dry without wiping it off?

Answer

Fisheyes, small round craters where residual contamination repels the paint.

#112 · Prepare Vehicle for Refinishing

At what two stages is a panel typically degreased?

Answer

Before sanding begins and again after sanding is completed.

#113 · Prepare Vehicle for Refinishing

What is a tack cloth used for?

Answer

A slightly sticky cloth used as the final wipe to lift fine dust without leaving residue.

#114 · Prepare Vehicle for Refinishing

What is the main job of primer?

Answer

To promote adhesion between the substrate and topcoat and build up minor surface imperfections for leveling.

#115 · Prepare Vehicle for Refinishing

When is epoxy primer typically the best choice?

Answer

Directly over sanded bare steel or aluminum, especially in corrosion prone areas, because it bonds strongly and resists moisture.

#116 · Prepare Vehicle for Refinishing

What is urethane primer surfacer designed to do?

Answer

Fill small sand scratches, pinholes, and surface irregularities so the area can be block sanded flat.

#117 · Prepare Vehicle for Refinishing

What is self-etch primer and when is it used?

Answer

A thin, fast drying primer with a mild acid that chemically etches bare metal, often used on galvanized steel or aluminum.

#118 · Prepare Vehicle for Refinishing

Why doesn't self-etch primer replace urethane surfacer for heavy filling?

Answer

Self-etch primer is thin and is not designed to build material the way a surfacer does.

#119 · Prepare Vehicle for Refinishing

What risk comes from spraying heavy urethane surfacer directly over bare steel with no epoxy or etch step first?

Answer

Weaker adhesion and increased corrosion risk in moisture prone areas.

#120 · Prepare Vehicle for Refinishing

What is the main job of sealer applied right before color?

Answer

To create a uniform, non-porous surface with consistent color and texture across the whole panel.

#121 · Prepare Vehicle for Refinishing

What problem does sealer help prevent from developing under a color coat?

Answer

Sand scratch swelling, where scratches soak up solvent from wet paint and show through the finish.

#122 · Prepare Vehicle for Refinishing

Why is sealer often tinted close to the topcoat color?

Answer

It improves hiding and reduces the number of color coats needed for full, even coverage.

#123 · Prepare Vehicle for Refinishing

How many coats of sealer are typically applied before color?

Answer

One or two light, even coats, followed by a short flash time per the technical data sheet.

#124 · Prepare Vehicle for Refinishing

What is the recommended gun to panel distance range for most spray guns?

Answer

Around six to eight inches from the surface.

#125 · Prepare Vehicle for Refinishing

What happens if the spray gun is held too far from the panel?

Answer

Dry spray results, with poor gloss and poor adhesion, sometimes developing into orange peel.

#126 · Prepare Vehicle for Refinishing

What happens if the spray gun is held too close to the panel?

Answer

Runs and sags form as too much wet paint lands in one spot too fast.

#127 · Prepare Vehicle for Refinishing

What happens if the gun is tilted instead of held perpendicular?

Answer

Uneven mil buildup, with more coating thickness on one side of the pattern than the other.

#128 · Prepare Vehicle for Refinishing

What happens if air pressure is set too low for the paint being sprayed?

Answer

The paint is not properly atomized, leaving coarse droplets and a rough, gritty finish.

#129 · Prepare Vehicle for Refinishing

What happens if air pressure is set too high for the paint being sprayed?

Answer

The paint is over atomized, causing excess overspray, dry edges, and reduced transfer efficiency.

#130 · Prepare Vehicle for Refinishing

What is a vertical fan pattern typically used for?

Answer

Spraying vertical panels like doors and quarter panels in horizontal, overlapping passes.

#131 · Prepare Vehicle for Refinishing

What is a round spray pattern typically used for?

Answer

Small spot repairs, edges, and hard to reach corners rather than large open panels.

#132 · Prepare Vehicle for Refinishing

What does the spray booth's filtered airflow protect against?

Answer

Dust and contamination reaching wet paint while it is applied and while it cures.

#133 · Prepare Vehicle for Refinishing

What happens if the spray booth is too cold when spraying?

Answer

Solvent evaporation slows down, increasing the risk of runs, sags, and extended dry times.

#134 · Prepare Vehicle for Refinishing

What happens if the spray booth is too hot when spraying?

Answer

Solvents flash off too fast and can trap solvent under a skinned over surface, causing a rough or textured finish.

#135 · Prepare Vehicle for Refinishing

What finish defect can high booth humidity cause?

Answer

A hazy or blushing finish where the film looks cloudy instead of clear and glossy.

#136 · Prepare Vehicle for Refinishing

What problem can a clogged booth intake filter cause?

Answer

Reduced airflow, starving the booth of the steady filtered air needed to keep dust off the wet finish.

#137 · Prepare Vehicle for Refinishing

What is a guide coat and how is it used?

Answer

A light, contrasting dusting sprayed over primer that sands away first from high spots, revealing low spots that need more leveling.

#138 · Prepare Vehicle for Refinishing

What hands-on check confirms a featheredge and surrounding panel are ready for color?

Answer

Running a clean, flat hand across the area to feel for any transition, ridge, or texture difference.

#139 · Prepare Vehicle for Refinishing

What must be confirmed about primer and sealer before color is sprayed?

Answer

That they have fully cured or flashed according to the product's technical data sheet.

#140 · Prepare Vehicle for Refinishing

What is a comeback in an auto body shop?

Answer

A vehicle that must return to the shop because the finished paint job failed inspection or developed a visible defect.

#141 · Repair Trim Components and Hardware

What is a Christmas tree clip?

Answer

A ribbed plastic push pin fastener that snaps into a pre-drilled hole to hold trim flush against the body.

#142 · Repair Trim Components and Hardware

What tool removes push clips and pries trim panels without scratching paint?

Answer

A trim panel removal tool, a flat plastic pry tool made for this purpose.

#143 · Repair Trim Components and Hardware

What rubber compound is standard for modern weatherstripping?

Answer

EPDM, ethylene propylene diene monomer, chosen for its resistance to sun and heat cracking.

#144 · Repair Trim Components and Hardware

What flexible plastic is used to mold bumper covers and body side mouldings?

Answer

TPO, thermoplastic olefin, which flexes on light impact instead of cracking.

#145 · Repair Trim Components and Hardware

Why can't scratched chrome trim usually be sanded and repainted?

Answer

Chrome is a thin plated finish, not paint, so sanding damages the plating instead of smoothing it.

#146 · Repair Trim Components and Hardware

What is a sems screw?

Answer

A screw with a permanently attached captive washer, used for a wider bearing surface on soft plastic.

#147 · Repair Trim Components and Hardware

What is a self-tapping screw used for in trim work?

Answer

It cuts its own mating threads into plastic or thin metal, common in mirror housings.

#148 · Repair Trim Components and Hardware

What is a snap clip?

Answer

A plastic clip molded onto the trim backing that flexes inward and springs out to lock behind a panel.

#149 · Repair Trim Components and Hardware

Why are snap clips treated as single-use fasteners?

Answer

They flex during install and can weaken or crack once released, risking a rattle if reused.

#150 · Repair Trim Components and Hardware

What is a barbed clip?

Answer

A fastener with angled ridges along its shaft that catch and resist being pulled back out.

#151 · Repair Trim Components and Hardware

Where are barbed clips most commonly used?

Answer

In door panel attachment points, where they are usually destroyed during removal and need replacing.

#152 · Repair Trim Components and Hardware

What is the paper strip test used for?

Answer

Checking weatherstripping compression by closing a strip of paper in the door and pulling it out to feel resistance.

#153 · Repair Trim Components and Hardware

What does the cowl panel cover?

Answer

The wiper linkage assembly and the cabin fresh air intake at the base of the windshield.

#154 · Repair Trim Components and Hardware

What holds a cowl panel in place?

Answer

A mix of plastic push clips and a small number of retaining screws.

#155 · Repair Trim Components and Hardware

What is the wiper linkage?

Answer

A set of connected arms that transfers motion from a single wiper motor to both wiper arms.

#156 · Repair Trim Components and Hardware

What commonly bends a wiper linkage arm?

Answer

Forcing a frozen wiper blade free from an icy windshield.

#157 · Repair Trim Components and Hardware

What does a bent wiper linkage cause?

Answer

Uneven wiper movement, stalling partway through the sweep, or grinding noise.

#158 · Repair Trim Components and Hardware

What should be checked when a cowl seal at the windshield base looks pinched?

Answer

That it sits flat and unpinched, or it can let water into the wiper motor or cabin intake.

#159 · Repair Trim Components and Hardware

What is the purpose of weatherstripping around a door opening?

Answer

To seal out wind noise, water intrusion, dust, and exhaust fumes.

#160 · Repair Trim Components and Hardware

What causes a door to sag over time?

Answer

Worn hinge pins or worn hinge bushings that no longer hold tightly.

#161 · Repair Trim Components and Hardware

What symptom shows a door is sagging from hinge wear?

Answer

Scraping against the fender when opening or uneven panel gaps.

#162 · Repair Trim Components and Hardware

What repair avoids full hinge replacement when the hinge body is still sound?

Answer

A hinge pin and bushing rebuild kit.

#163 · Repair Trim Components and Hardware

What does a striker plate do?

Answer

It is mounted on the body pillar and catches the door latch to hold the door shut.

#164 · Repair Trim Components and Hardware

What is the first check for a door that needs slamming to latch?

Answer

Whether the striker plate needs a slight adjustment before assuming the latch is broken.

#165 · Repair Trim Components and Hardware

How is a striker plate typically adjusted?

Answer

By loosening its mounting screws and shifting its position slightly before retightening.

#166 · Repair Trim Components and Hardware

What is the correct diagnostic step for a loose door handle that won't release the latch?

Answer

Open the door panel and watch the linkage move as the handle is operated.

#167 · Repair Trim Components and Hardware

What connects a door handle to the latch mechanism?

Answer

A rod or cable linkage.

#168 · Repair Trim Components and Hardware

What material typically forms the rigid backing of interior door panels?

Answer

Injection molded plastic, often covered with a soft skin, foam, or fabric layer.

#169 · Repair Trim Components and Hardware

What emblem removal technique avoids scratching paint?

Answer

Working dental floss or fishing line behind the emblem to cut the old adhesive.

#170 · Repair Trim Components and Hardware

What should be aligned first when installing a new emblem?

Answer

The locating pins with their corresponding holes in the body panel.

#171 · Repair Trim Components and Hardware

Why does an emblem need a specified curing time before a car wash?

Answer

The adhesive needs that time to fully bond, or the emblem risks coming loose again.

#172 · Repair Trim Components and Hardware

How is old emblem adhesive residue safely softened for removal?

Answer

Gentle, moving heat from a heat gun followed by an approved adhesive remover.

#173 · Repair Trim Components and Hardware

What should be checked before installing a new mirror housing after a collision?

Answer

Whether the underlying mounting bracket itself is bent, cracked, or broken.

#174 · Repair Trim Components and Hardware

How is mirror glass typically removed from a power mirror actuator?

Answer

By carefully prying it free of its small retaining spring clips.

#175 · Repair Trim Components and Hardware

What should be tested on a mirror before the door panel is fully reassembled?

Answer

Its full adjustment range of motion and its heating function if equipped.

#176 · Repair Trim Components and Hardware

Why match the original fastener type and size exactly during trim reassembly?

Answer

A mismatched fastener can strip the mounting hole or fail to seal the trim properly.

#177 · Repair Trim Components and Hardware

What is the two-sided automotive molding tape designed to withstand?

Answer

Vibration and temperature swings without releasing, unlike household double-sided tape.

#178 · Repair Trim Components and Hardware

What unit measures panel gap specifications?

Answer

Millimetres.

#179 · Repair Trim Components and Hardware

What confirms a repaired panel gap is finished correctly?

Answer

The measured gaps match factory specification and stay even top to bottom.

#180 · Repair Trim Components and Hardware

What should a technician do with hidden damage found behind an intact-looking moulding?

Answer

Photograph it and add it to the estimate through a supplement.

#181 · Repair Trim Components and Hardware

What is the best reference for hidden fastener locations before disassembly?

Answer

The manufacturer's official service information for that exact vehicle model.

#182 · Repair Trim Components and Hardware

What usually happens if a stuck trim panel is pulled without locating every hidden clip?

Answer

A mounting tab snaps off and the panel needs replacement.

#183 · Repair Trim Components and Hardware

Why should a technician clean the channel before installing new weatherstripping?

Answer

Leftover adhesive or debris can prevent the new seal from seating fully.

#184 · Repair Trim Components and Hardware

What confirms a weatherstrip replacement is properly seated all the way around?

Answer

Repeating the paper strip test at multiple points around the full perimeter.

#185 · Repair Trim Components and Hardware

What is the first step in isolating a rattle from a repaired door panel?

Answer

Trace the rattle back to a specific fastener, clip, or seal that was not fully seated, then correct it.

#186 · Repair Trim Components and Hardware

What is the most likely cause of new wind noise after a trim reinstallation?

Answer

A fastener, clip, or seal somewhere that was not fully seated during reassembly.

#187 · Repair Trim Components and Hardware

What comparison helps catch a trim piece sitting slightly wrong during final inspection?

Answer

Comparing the repaired side to the same undamaged area on the opposite side.

#188 · Repair Trim Components and Hardware

What check confirms a latch and striker repair beyond a visual inspection?

Answer

Closing the door several times and confirming a consistent, easy latch.

#189 · Repair Trim Components and Hardware

What causes stripped plastic threads when reinstalling a self-tapping screw in a mirror housing?

Answer

Overtightening the screw well beyond a firm, properly snug fit.

#190 · Repair Trim Components and Hardware

What is the correct action for a broken cowl panel clip found during removal?

Answer

Order and install a replacement clip of the correct original type.

#191 · Inspect Remove and Replace Supplemental Restraint Systems

What does SRS stand for and what does the word supplemental mean in this context?

Answer

SRS stands for Supplemental Restraint System. Supplemental means the airbags and pretensioners work together with the seatbelt as backup protection, not as a replacement for it.

#192 · Inspect Remove and Replace Supplemental Restraint Systems

Name the five main components that make up a typical SRS.

Answer

Airbag modules, seatbelt pretensioners, impact sensors, the clockspring, and the SRS control module. Every part reports to or is controlled by the central module.

#193 · Inspect Remove and Replace Supplemental Restraint Systems

What does a seatbelt pretensioner do and when does it fire relative to the airbag?

Answer

It pulls slack out of the seatbelt in the first instant of a crash, before the airbag inflates, holding the occupant firmly against the seat.

#194 · Inspect Remove and Replace Supplemental Restraint Systems

Can a fired seatbelt pretensioner be reset and reused?

Answer

No. Pretensioners are one-time-use pyrotechnic devices. Once fired, both the pretensioner and the belt must be replaced.

#195 · Inspect Remove and Replace Supplemental Restraint Systems

What job do impact sensors perform in the SRS?

Answer

They detect a sudden change in vehicle speed matching a real collision pattern and send that signal to the SRS control module to decide which devices to fire.

#196 · Inspect Remove and Replace Supplemental Restraint Systems

What is the clockspring and why is it needed behind the steering wheel?

Answer

It is a coiled ribbon of wiring that carries electrical signals from the fixed steering column to the rotating wheel, allowing the driver airbag and horn to keep working through full rotations.

#197 · Inspect Remove and Replace Supplemental Restraint Systems

How should the SRS warning light behave during a normal start-up with no faults?

Answer

It comes on briefly for a few seconds and then turns off, showing the self-test completed with no stored faults found.

#198 · Inspect Remove and Replace Supplemental Restraint Systems

What should a technician do first when the SRS warning light stays on or flashes while driving?

Answer

Connect a scan tool capable of reading SRS-specific diagnostic trouble codes, rather than guessing at the cause without pulling codes.

#199 · Inspect Remove and Replace Supplemental Restraint Systems

Why should a technician check for stored SRS codes even after the warning light goes off?

Answer

Some codes stay logged in the control module's memory even after the light stops flashing on a given key cycle, so they must be checked and cleared with a scan tool.

#200 · Inspect Remove and Replace Supplemental Restraint Systems

What is the very first step in safely de-arming an SRS before any component work begins?

Answer

Disconnect the vehicle battery, starting with the negative, or ground, cable, so no current can reach the airbag circuits.

#201 · Inspect Remove and Replace Supplemental Restraint Systems

Why is the SRS not automatically safe the instant the battery cable is disconnected?

Answer

A backup power capacitor inside the SRS control module still holds enough charge to fire the airbags for a short time after disconnection.

#202 · Inspect Remove and Replace Supplemental Restraint Systems

How long should a technician wait after disconnecting the battery before touching SRS connectors?

Answer

Always the OEM specified discharge time for that exact vehicle, confirmed from the manufacturer's service information. This time varies by manufacturer, so never assume one universal number.

#203 · Inspect Remove and Replace Supplemental Restraint Systems

What colour are SRS wiring harness connectors usually manufactured in, and why?

Answer

Yellow, so they stand out clearly from ordinary vehicle wiring and help technicians avoid accidental damage or disconnection.

#204 · Inspect Remove and Replace Supplemental Restraint Systems

Why must a secondary locking clip on an SRS connector always be fully engaged during reinstallation?

Answer

A connector that looks connected but is not fully locked can cause an intermittent fault or a failure to deploy in a real crash.

#205 · Inspect Remove and Replace Supplemental Restraint Systems

Why should a standard multimeter never be used to test directly across an inflator's terminals with power applied?

Answer

It takes very little current to fire an inflator, and a standard multimeter can pass enough current through its leads to trigger deployment.

#206 · Inspect Remove and Replace Supplemental Restraint Systems

How should a removed airbag module be carried and set down, and why?

Answer

Trim cover facing up and away from the technician's body, so any unexpected deployment force is directed upward and outward rather than toward a person.

#207 · Inspect Remove and Replace Supplemental Restraint Systems

Where should a removed airbag module never be placed while waiting for reinstallation?

Answer

Never face down, stacked under other parts, or leaned so the cover faces a walking path. It should sit on a stable flat surface, cover up, clear of foot traffic.

#208 · Inspect Remove and Replace Supplemental Restraint Systems

What must happen to a deployed airbag module or a fired pretensioner after removal?

Answer

It must be disposed of as hazardous material according to shop safety rules. It can never be reused, rebuilt, or installed on another vehicle.

#209 · Inspect Remove and Replace Supplemental Restraint Systems

What is the correct order for removing an airbag module from the steering wheel or dash?

Answer

Access the module through the trim panel, disconnect its electrical connector, then remove the mounting fasteners so the module comes free.

#210 · Inspect Remove and Replace Supplemental Restraint Systems

Why must a replacement SRS part be the exact OEM part or an approved equivalent for that specific trim level?

Answer

SRS parts are matched to that vehicle's crash sensing calibration. A part from a different trim can change how the system fires even if it physically fits.

#211 · Inspect Remove and Replace Supplemental Restraint Systems

What tool must be used to tighten SRS mounting fasteners, and why not just tighten by feel?

Answer

A calibrated torque wrench set to the exact specification. Under-torquing can let a module shift loose, and over-torquing can crack the housing or mounting boss.

#212 · Inspect Remove and Replace Supplemental Restraint Systems

What can happen if a steering wheel is rotated while off the column and the clockspring is not centered and locked?

Answer

The ribbon inside can overwind or underwind, which can tear it or leave the driver airbag circuit open so the airbag will not fire.

#213 · Inspect Remove and Replace Supplemental Restraint Systems

What must a technician confirm about the clockspring before bolting the steering wheel back onto the column?

Answer

That the clockspring is centered, following the specific centering procedure listed for that vehicle in the service manual.

#214 · Inspect Remove and Replace Supplemental Restraint Systems

What two checks make up final verification after finishing SRS reinstallation work?

Answer

Confirm the SRS warning light completes its self-test and turns off, and use a scan tool to check for and clear any stored SRS trouble codes.

#215 · Inspect Remove and Replace Supplemental Restraint Systems

Besides the warning light and scan tool check, what else should be visually confirmed before releasing an SRS repair to the customer?

Answer

That no wiring is pinched under new trim or panels, every connector is fully seated with its secondary lock engaged, and any relocated sensor is mounted exactly where specified.

#216 · Weld Heat and Cut MIG and Oxyfuel

What does MIG stand for?

Answer

Metal inert gas, the shielding gas welding process used for most auto body sheet metal repairs.

#217 · Weld Heat and Cut MIG and Oxyfuel

What wire diameter is standard for thin body sheet metal?

Answer

0.023 or 0.024 inch wire, which produces a smaller, cooler puddle that resists burn-through.

#218 · Weld Heat and Cut MIG and Oxyfuel

What wire diameter suits thicker structural steel?

Answer

0.030 inch wire, used for frame rails and other heavy structural components.

#219 · Weld Heat and Cut MIG and Oxyfuel

What shielding gas blend is standard for mild steel body panels?

Answer

75 percent argon and 25 percent carbon dioxide, giving a smooth arc and low spatter.

#220 · Weld Heat and Cut MIG and Oxyfuel

What is the standard MIG gas flow rate for mild steel?

Answer

20 to 25 cubic feet per hour.

#221 · Weld Heat and Cut MIG and Oxyfuel

What happens if MIG shielding gas flow is too low?

Answer

Outside air contaminates the puddle, which can cause porosity in the finished weld.

#222 · Weld Heat and Cut MIG and Oxyfuel

What happens if MIG shielding gas flow is too high?

Answer

Turbulence can pull outside air back into the shielding zone and disrupt coverage.

#223 · Weld Heat and Cut MIG and Oxyfuel

What does a correctly tuned MIG arc sound like?

Answer

Steady frying bacon, indicating stable arc length and consistent metal transfer.

#224 · Weld Heat and Cut MIG and Oxyfuel

What does a popping, crackling MIG arc sound usually mean?

Answer

Voltage too low or wire speed too fast, causing the wire to stub into the metal.

#225 · Weld Heat and Cut MIG and Oxyfuel

What does a hissing, spattering MIG arc with a wide puddle usually mean?

Answer

Voltage set too high for the material being welded.

#226 · Weld Heat and Cut MIG and Oxyfuel

Define a butt joint.

Answer

Two panel edges placed directly against each other with no overlap, with weld metal filling the gap.

#227 · Weld Heat and Cut MIG and Oxyfuel

Define a lap joint.

Answer

One piece of metal overlapping another, with the weld run along the overlapping edge.

#228 · Weld Heat and Cut MIG and Oxyfuel

Define a fillet weld.

Answer

A weld joining two pieces of metal meeting at an angle, forming a triangular cross section.

#229 · Weld Heat and Cut MIG and Oxyfuel

Define a plug weld.

Answer

A weld made through a round hole in the top panel, fusing it to the panel beneath.

#230 · Weld Heat and Cut MIG and Oxyfuel

What do plug welds replace from factory assembly?

Answer

The spot welds used to join panels during original vehicle assembly.

#231 · Weld Heat and Cut MIG and Oxyfuel

Why are butt joints used for panel splices?

Answer

They keep the original panel contour without adding extra thickness to the repair area.

#232 · Weld Heat and Cut MIG and Oxyfuel

Why are lap joints easier to fit up than butt joints?

Answer

They do not require a precise edge to edge gap match before welding.

#233 · Weld Heat and Cut MIG and Oxyfuel

What does a weld symbol below the reference line mean?

Answer

The weld goes on the arrow side, the side the arrow physically touches.

#234 · Weld Heat and Cut MIG and Oxyfuel

What does a weld symbol above the reference line mean?

Answer

The weld goes on the side opposite the arrow.

#235 · Weld Heat and Cut MIG and Oxyfuel

What does a symbol on both sides of the reference line mean?

Answer

Welds are required on both sides of the joint.

#236 · Weld Heat and Cut MIG and Oxyfuel

What does a number to the left of a weld symbol usually give?

Answer

The weld size, such as the leg length of a fillet weld in millimeters.

#237 · Weld Heat and Cut MIG and Oxyfuel

What does a length and pitch number pair to the right of a symbol give?

Answer

The length of each weld segment and the spacing between segments for an intermittent weld.

#238 · Weld Heat and Cut MIG and Oxyfuel

What does a circle at the symbol bend indicate?

Answer

The weld must go all the way around the joint.

#239 · Weld Heat and Cut MIG and Oxyfuel

What does a flag at the symbol bend indicate?

Answer

A field weld, made on site rather than in a factory jig.

#240 · Weld Heat and Cut MIG and Oxyfuel

Why must technicians read weld symbols accurately?

Answer

Manufacturer procedures specify exact joint type, size, and location using this symbol system.

#241 · Weld Heat and Cut MIG and Oxyfuel

Define porosity in a MIG weld.

Answer

Small round holes or pits caused by gas bubbles trapped in the cooling metal.

#242 · Weld Heat and Cut MIG and Oxyfuel

What commonly causes porosity?

Answer

Contamination such as rust, paint, oil, or moisture, or disrupted shielding gas flow.

#243 · Weld Heat and Cut MIG and Oxyfuel

Define burn-through.

Answer

The arc melting completely through thin sheet metal, leaving a hole instead of a bead.

#244 · Weld Heat and Cut MIG and Oxyfuel

What commonly causes burn-through?

Answer

Excess heat for the material thickness, such as voltage too high or dwelling too long.

#245 · Weld Heat and Cut MIG and Oxyfuel

Define cold lap.

Answer

Weld metal that flows over the base metal surface without truly fusing into it.

#246 · Weld Heat and Cut MIG and Oxyfuel

What commonly causes cold lap?

Answer

Voltage too low, travel speed too fast, or a dirty joint surface.

#247 · Weld Heat and Cut MIG and Oxyfuel

Define undercut.

Answer

A groove melted into the base metal beside the bead that is not filled back in.

#248 · Weld Heat and Cut MIG and Oxyfuel

What commonly causes undercut?

Answer

Excess heat combined with an incorrect gun or torch angle during the weld pass.

#249 · Weld Heat and Cut MIG and Oxyfuel

What commonly causes excessive spatter?

Answer

Voltage too high, incorrect gas mix, or a dirty, worn contact tip.

#250 · Weld Heat and Cut MIG and Oxyfuel

What fuel gas is most common in auto body oxyfuel work?

Answer

Acetylene, paired with oxygen to produce a flame hot enough for heating and cutting steel.

#251 · Weld Heat and Cut MIG and Oxyfuel

Why must acetylene cylinders stay upright?

Answer

Acetone inside the cylinder stabilizes the gas and can be drawn into hoses if tipped.

#252 · Weld Heat and Cut MIG and Oxyfuel

What method checks oxyfuel hose connections for leaks?

Answer

A soap solution brushed onto the connections, since escaping gas forms visible bubbles.

#253 · Weld Heat and Cut MIG and Oxyfuel

What is the correct cylinder valve opening sequence for lighting?

Answer

Open the oxygen cylinder valve fully, then the acetylene cylinder valve about three quarters turn.

#254 · Weld Heat and Cut MIG and Oxyfuel

What tool safely ignites an oxyfuel torch?

Answer

A friction lighter, which keeps the ignition source at a safe distance from the tip.

#255 · Weld Heat and Cut MIG and Oxyfuel

Describe a neutral flame.

Answer

A flame with a well-defined inner cone and no feather of extra acetylene trailing off.

#256 · Weld Heat and Cut MIG and Oxyfuel

What is the first valve closed during oxyfuel shutdown?

Answer

The torch acetylene valve, closed first to extinguish the flame.

#257 · Weld Heat and Cut MIG and Oxyfuel

What is the full oxyfuel shutdown order after the flame is out?

Answer

Close the torch oxygen valve, close both cylinder valves, then bleed the hoses at the torch.

#258 · Weld Heat and Cut MIG and Oxyfuel

Why bleed the hoses during shutdown?

Answer

To release remaining pressure from the lines and prevent a flashback risk next use.

#259 · Weld Heat and Cut MIG and Oxyfuel

What color signals correct shrinking temperature for mild steel?

Answer

A dull cherry red, roughly 1400 to 1600 degrees Fahrenheit.

#260 · Weld Heat and Cut MIG and Oxyfuel

Why does a heated spot thicken instead of spreading during shrinking?

Answer

Surrounding cooler metal restrains the expansion, forcing the hot spot to thicken.

#261 · Weld Heat and Cut MIG and Oxyfuel

How is a heated spot cooled during panel shrinking?

Answer

With a wet rag and light hammer and dolly pressure, shrinking it tighter as it cools.

#262 · Weld Heat and Cut MIG and Oxyfuel

Why is shrinking done in small spots rather than one large area?

Answer

Working from the outside in with small spots avoids overheating and burning the metal.

#263 · Weld Heat and Cut MIG and Oxyfuel

What risk comes from overheating a panel during shrinking?

Answer

Burning the metal's grain structure, leaving it weaker and more brittle than before.

#264 · Weld Heat and Cut MIG and Oxyfuel

What is the kindling temperature steel must reach before cutting?

Answer

Roughly 1600 degrees Fahrenheit, reached by the torch's outer preheating flames.

#265 · Weld Heat and Cut MIG and Oxyfuel

What does the cutting oxygen jet do once steel is preheated?

Answer

It burns through the hot steel and blows the resulting oxide out of the cut.

#266 · Weld Heat and Cut MIG and Oxyfuel

Define kerf.

Answer

The width of material removed along a cutting line.

#267 · Weld Heat and Cut MIG and Oxyfuel

What happens if oxyfuel cutting travel speed is too fast?

Answer

Unmelted metal remains below the surface and the cut does not fully separate.

#268 · Weld Heat and Cut MIG and Oxyfuel

What happens if oxyfuel cutting travel speed is too slow?

Answer

The kerf widens, gas is wasted, and heat distortion in the panel increases.

#269 · Weld Heat and Cut MIG and Oxyfuel

What tip height is typical for oxyfuel cutting?

Answer

About one eighth to three sixteenths of an inch above the metal surface.

#270 · Weld Heat and Cut MIG and Oxyfuel

Why is oxyfuel cutting mainly used for rough removal work?

Answer

The heat distorts surrounding metal and leaves a rough edge that needs grinding.

#271 · Weld Heat and Cut MIG and Oxyfuel

What restriction applies to cutting torches near modern vehicle structures?

Answer

Manufacturers restrict cutting near high strength steel, adhesive joints, or airbag components.

#272 · Weld Heat and Cut MIG and Oxyfuel

Why is a welding helmet used for MIG but goggles for oxyfuel?

Answer

MIG arc flash is ultraviolet and intense, while torch flame gives strong visible light only.

#273 · Weld Heat and Cut MIG and Oxyfuel

Why avoid synthetic fabrics like polyester when welding or cutting?

Answer

Sparks can melt synthetic fabric onto the skin, causing burns.

#274 · Weld Heat and Cut MIG and Oxyfuel

What clearance distance from flammables is recommended for hot work?

Answer

At least 35 feet whenever practical.

#275 · Weld Heat and Cut MIG and Oxyfuel

Why check the opposite side of a panel before heating or cutting?

Answer

Sparks and heat can pass through thin metal and ignite hidden flammable material.

#276 · Weld Heat and Cut MIG and Oxyfuel

Why is ventilation important during extended MIG or oxyfuel work?

Answer

Coating fumes, oxygen depletion, and carbon monoxide buildup can become health hazards.

#277 · Weld Heat and Cut MIG and Oxyfuel

What does a completed weld need for visual inspection?

Answer

Uniform bead width, full fusion into the base metal, and no cracks, porosity, or undercut.

#278 · Weld Heat and Cut MIG and Oxyfuel

What is a destructive test coupon?

Answer

A scrap piece welded the same way as the repair, then bent or broken to check strength.

#279 · Weld Heat and Cut MIG and Oxyfuel

What does a strong weld coupon do when bent or broken?

Answer

It tears through the surrounding base metal rather than breaking along the weld line.

#280 · Weld Heat and Cut MIG and Oxyfuel

What does a coupon breaking cleanly along the weld line indicate?

Answer

The weld did not achieve adequate fusion strength.

#281 · Weld Heat and Cut MIG and Oxyfuel

Why apply corrosion protection to bare metal after welding?

Answer

To prevent rust from starting inside the repaired structure where it cannot be seen.

#282 · Weld Heat and Cut MIG and Oxyfuel

Why must manufacturer joint type and location be followed exactly?

Answer

Manufacturers engineer specific joint strengths into the vehicle for crash performance.

#283 · Weld Heat and Cut MIG and Oxyfuel

What voltage and wire speed trend applies to thin sheet metal?

Answer

Lower voltage and slower wire speed than thicker structural steel requires.

#284 · Weld Heat and Cut MIG and Oxyfuel

What is the purpose of a welder's inside cover setting chart?

Answer

It gives a starting voltage and wire speed to fine tune from during setup.

#285 · Weld Heat and Cut MIG and Oxyfuel

Why does galvanized steel produce more fumes and spatter when welded?

Answer

The zinc coating burns off during welding, producing strong fumes and extra spatter.

#286 · Weld Heat and Cut MIG and Oxyfuel

What does a loose MIG ground clamp connection cause?

Answer

An unstable arc, possible stray arc strikes, and poor weld quality.

#287 · Weld Heat and Cut MIG and Oxyfuel

What does a V-groove weld symbol require in edge preparation?

Answer

A beveled or grooved edge so the weld can fill the groove and gain penetration.

#288 · Weld Heat and Cut MIG and Oxyfuel

Why is a square butt joint limited mainly to thin material?

Answer

Without a bevel, full penetration is difficult to achieve reliably on thicker steel.

#289 · Weld Heat and Cut MIG and Oxyfuel

What does a T-bevel joint provide compared to a standard T-fillet joint?

Answer

Deeper penetration into the joint, useful on thicker material at a right angle intersection.

#290 · Weld Heat and Cut MIG and Oxyfuel

What is the total included angle of a V-butt joint with 30 degree bevels on each side?

Answer

60 degrees, the sum of the bevel angle on each side of the joint.

#291 · Repair Vehicle Bodies and Components

What is direct damage in a collision repair context?

Answer

Direct damage is the crush, tear, or crease found right at the point where the vehicle was struck during the collision.

#292 · Repair Vehicle Bodies and Components

What is indirect damage?

Answer

Indirect damage is damage that shows up away from the impact point because collision force traveled through the frame or unibody structure.

#293 · Repair Vehicle Bodies and Components

What is unibody construction?

Answer

Unibody construction welds the floor pan, rails, and pillars into one connected structural shell instead of using a separate bolt-on frame.

#294 · Repair Vehicle Bodies and Components

Why do technicians check gaps on the undamaged side of a unibody vehicle?

Answer

Because a hit on one corner can twist or stretch metal on the opposite side through the connected structure, so a shift there confirms hidden damage.

#295 · Repair Vehicle Bodies and Components

What are diagonal measurements used for?

Answer

Diagonal measurements compare fixed reference points on one side of the vehicle to mirror points on the other side to detect hidden structural shift.

#296 · Repair Vehicle Bodies and Components

What does unequal diagonal measurement readings indicate?

Answer

Unequal diagonal readings indicate the vehicle's structure has shifted out of its original shape, even if the sheet metal looks fine to the eye.

#297 · Repair Vehicle Bodies and Components

What does cracked paint at the center of a dent usually indicate?

Answer

Cracked or flaking paint at a dent's center usually means the metal stretched past its elastic limit rather than simply bending.

#298 · Repair Vehicle Bodies and Components

What does intact paint on a dent usually indicate?

Answer

Intact paint with no cracking usually means the metal only bent and will spring back more easily, making for an easier repair.

#299 · Repair Vehicle Bodies and Components

What is crash mechanism, and why does it matter?

Answer

Crash mechanism is how, from what angle, and at roughly what speed a vehicle was struck; knowing it helps predict where stress traveled and hidden damage may exist.

#300 · Repair Vehicle Bodies and Components

What is shrinking in metal repair?

Answer

Shrinking removes excess stretched metal by heating a small spot and rapidly cooling it under light hammer pressure, pulling surrounding metal tighter.

#301 · Repair Vehicle Bodies and Components

What is oil canning?

Answer

Oil canning is a panel that flexes in and out like the bottom of an oil can because it has lost its original crown from stretched metal.

#302 · Repair Vehicle Bodies and Components

What is a dolly used for in body repair?

Answer

A dolly is a shaped piece of hardened steel held against the backside of a panel to act as an anvil while the front is hammered.

#303 · Repair Vehicle Bodies and Components

What is the difference between hammer on dolly and hammer off dolly?

Answer

Hammer on dolly strikes directly over the dolly for fine finishing on nearly straight metal; hammer off dolly strikes slightly away to move larger high and low spots.

#304 · Repair Vehicle Bodies and Components

What is a stud welder used for?

Answer

A stud welder attaches small studs, pins, or washers to a damaged panel with a short weld so a slide hammer can pull out the dent.

#305 · Repair Vehicle Bodies and Components

When is stud welding and pulling the preferred repair method?

Answer

It is preferred when a technician cannot reach the backside of a panel with a dolly, such as on rocker panels, quarter panels, or roof sections.

#306 · Repair Vehicle Bodies and Components

What is the correct working order when straightening a dent?

Answer

Work from the outer edge of the dent toward the center, since pulling or hammering the deepest point first stretches the metal further.

#307 · Repair Vehicle Bodies and Components

What is a vacuum cup puller used for?

Answer

A vacuum cup puller lifts shallow dents on flexible panels without heat or welding, useful for hail damage or small dings with intact paint.

#308 · Repair Vehicle Bodies and Components

What is body filler?

Answer

Body filler, also called plastic filler, is a two part material mixed with a hardener and applied over straightened metal to restore a smooth true contour.

#309 · Repair Vehicle Bodies and Components

Why must bare metal be sanded before applying filler?

Answer

Bare metal must be sanded to remove rust, old paint, and contamination, because filler will not bond properly to a dirty or coated surface.

#310 · Repair Vehicle Bodies and Components

What is featheredging?

Answer

Featheredging is sanding the edge of a repair area to a gradual taper so filler and paint blend smoothly into the surrounding original paint.

#311 · Repair Vehicle Bodies and Components

Why is filler applied in thin layers instead of one thick layer?

Answer

Thin layers generate less heat while curing and are less likely to crack or shrink unevenly than one thick application.

#312 · Repair Vehicle Bodies and Components

What tool shapes cured filler before finer sanding begins?

Answer

A surform, sometimes called a cheese grater file, has a perforated grater blade used to shape cured filler quickly before finer sanding.

#313 · Repair Vehicle Bodies and Components

What is a guide coat?

Answer

A guide coat is a thin, contrasting colored dust coat of primer sprayed over sanded filler that wears away from high spots first, revealing low spots.

#314 · Repair Vehicle Bodies and Components

What is a skim coat?

Answer

A skim coat is a very thin final layer of filler applied to fill tiny pinholes or sand scratches left from block sanding.

#315 · Repair Vehicle Bodies and Components

What is sectioning?

Answer

Sectioning is a partial replacement technique where only the damaged portion of a panel is cut out and a matching new section is welded in.

#316 · Repair Vehicle Bodies and Components

Why do manufacturers specify approved sectioning locations?

Answer

Approved sectioning locations keep the new weld joint away from high stress areas so the repair does not compromise crash performance.

#317 · Repair Vehicle Bodies and Components

Why are high strength and ultra high strength steels poor candidates for heat straightening?

Answer

Heating these steels permanently changes their strength properties, weakening the vehicle's crash performance even if the panel looks straight afterward.

#318 · Repair Vehicle Bodies and Components

What are structural panels?

Answer

Structural panels are part of the vehicle's designed crash energy path, such as frame rails, rocker panels, and certain pillars.

#319 · Repair Vehicle Bodies and Components

What are non-structural or cosmetic panels?

Answer

Non-structural panels, like a bolt on fender or hood, do not carry crash loads and generally allow more repair flexibility, including full replacement.

#320 · Repair Vehicle Bodies and Components

Why might a rusted rocker panel be replaced instead of repaired?

Answer

If rust has already spread from the inside out, it is often better to replace the panel because unremoved rust will continue spreading under a repair.

#321 · Repair Vehicle Bodies and Components

What is weld through primer?

Answer

Weld through primer is a zinc rich coating applied to metal edges before welding, designed to survive the welding heat and still protect the seam.

#322 · Repair Vehicle Bodies and Components

What is seam sealer used for?

Answer

Seam sealer is a flexible material applied to panel joints to restore the factory water sealing and sound deadening the joint originally provided.

#323 · Repair Vehicle Bodies and Components

What is cavity wax?

Answer

Cavity wax, or corrosion resistant wax, is sprayed into enclosed structures like rocker panels and door bottoms where liquid paint cannot reach directly.

#324 · Repair Vehicle Bodies and Components

What must be reapplied after a repair disturbs the underside of a vehicle?

Answer

Any undercoating removed during repair to access damage or hardware must be reapplied to match the vehicle's original protection level.

#325 · Repair Vehicle Bodies and Components

What are panel gaps checked against?

Answer

Panel gaps are checked against the vehicle manufacturer's original specification to confirm neighboring panels line up evenly all the way around.

#326 · Repair Vehicle Bodies and Components

What does an uneven panel gap usually indicate?

Answer

An uneven panel gap usually signals improper alignment or hidden structural damage in the hinge, panel, or underlying structure.

#327 · Repair Vehicle Bodies and Components

What hardware allows small gap and fit adjustments?

Answer

Hinges, latches, and striker plates usually have slotted mounting holes that allow small adjustments to correct gap and flush fit.

#328 · Repair Vehicle Bodies and Components

What is flush fit?

Answer

Flush fit describes how evenly one panel's surface lines up with its neighbor's surface, with no edge sitting noticeably higher or lower.

#329 · Repair Vehicle Bodies and Components

What does it mean for a panel to sit proud?

Answer

Sitting proud means a panel surface is raised above the surrounding surface, which can fail inspection even if the gap width is correct.

#330 · Repair Vehicle Bodies and Components

Why are trim clips and weatherstripping installed only after alignment is confirmed?

Answer

Installing trim too early can mask a poor fit, so alignment, gap, and flush fit are confirmed correct before trim is reinstalled.

#331 · Repair Vehicle Bodies and Components

Why is a repaired vehicle opened and closed several times after final adjustment?

Answer

Opening and closing the panel confirms it operates smoothly through its full range of motion without rubbing or binding.

#332 · Repair Vehicle Bodies and Components

Why must exposed bare metal always receive corrosion protection?

Answer

Rust starts quickly once bare steel is exposed to air and moisture, so corrosion protection is a required step, not an optional one.

#333 · Repair Vehicle Bodies and Components

What is checked when remeasuring structure after a repair?

Answer

Post repair measurements are compared against the pre repair measurements and factory specifications to confirm the structure returned within tolerance.

#334 · Repair Vehicle Bodies and Components

Why is paint match inspected under natural or full spectrum lighting?

Answer

Fluorescent shop lighting can hide color and texture differences that become obvious once the vehicle is viewed outdoors under natural light.

#335 · Repair Vehicle Bodies and Components

What is orange peel texture?

Answer

Orange peel texture is a bumpy, uneven paint surface that inspectors check for during final finish quality verification.

#336 · Repair Vehicle Bodies and Components

Why are mechanical and electrical systems tested after a body repair?

Answer

Systems like window regulators, mirror motors, or bumper sensors disturbed during repair must be tested for proper operation before delivery.

#337 · Repair Vehicle Bodies and Components

What is checked during a post repair road test?

Answer

A road test checks for proper wheel alignment, absence of unusual noises, and correct operation of safety systems affected by the repair.

#338 · Repair Vehicle Bodies and Components

What should repair documentation include?

Answer

Repair documentation should record the damage found, the repair method used for each panel, parts replaced versus repaired, and confirming measurements.

#339 · Repair Vehicle Bodies and Components

What is a crown in body repair terminology?

Answer

Crown is the natural curve built into a panel during manufacturing, and restoring it is part of a properly finished repair.

#340 · Repair Vehicle Bodies and Components

What are high spots and low spots?

Answer

High spots and low spots are areas of a panel sitting above or below the intended contour after straightening, found by sight and hand.

#341 · Repair Vehicle Bodies and Components

What is work hardening?

Answer

Work hardening is metal becoming stiffer and more brittle from repeated striking or bending without heat relief, which can cause cracking.

#342 · Repair Vehicle Bodies and Components

What is a heat affected zone?

Answer

A heat affected zone is the area around a weld or shrinking spot where heat has changed the metal's properties without melting it directly.

#343 · Repair Vehicle Bodies and Components

Why do technicians alternate their striking pattern during hammer work?

Answer

Alternating strikes avoids repeatedly hitting the same spot, which helps prevent work hardening that could cause the metal to crack.

#344 · Repair Vehicle Bodies and Components

What Ontario exam code is the Auto Body Repairer certificate of qualification?

Answer

The Ontario Auto Body Repairer certificate of qualification is exam code 310Q, a provincial-only exam separate from Red Seal trade 310B.

#345 · Repair Vehicle Bodies and Components

What is a flat sanding block used to detect?

Answer

A flat sanding block dragged across a panel contacts high spots first, showing exactly where more sanding or shaping work is needed.

#346 · Repair Vehicle Bodies and Components

Why is hammer and dolly work performed from both sides of a panel at once?

Answer

The dolly supports the metal from behind while the hammer works the front, gradually bringing low spots up and high spots down together.

#347 · Repair Vehicle Bodies and Components

What visual clue helps distinguish a bent dent from a stretched dent?

Answer

Cracked paint suggests stretched metal needing shrinking or filler, while intact paint suggests bent metal that may spring back more easily.

#348 · Repair Vehicle Bodies and Components

Why is grinding stud marks flush a delicate step?

Answer

Grinding too deeply can thin the metal excessively and weaken the panel, so only the stud remnant should be removed carefully.

#349 · Repair Vehicle Bodies and Components

What happens if seam sealer does not match the factory bead pattern?

Answer

A mismatched bead can compromise water sealing and sound deadening and may signal inconsistent coverage that risks a future leak.

#350 · Repair Vehicle Bodies and Components

Why should drain holes remain clear when applying cavity wax?

Answer

Drain holes allow water inside a cavity to escape; blocking them traps moisture and can cause accelerated corrosion instead of protection.

#351 · Repair Vehicle Bodies and Components

What repair difference exists between aluminum and steel panels?

Answer

Aluminum has different heat sensitivity than steel and requires manufacturer specific repair methods and equipment rather than steel-based techniques.

#352 · Repair Vehicle Bodies and Components

Why is a test fit performed before painting a new replacement panel?

Answer

A test fit confirms gap and flush fit while the part can still be adjusted, avoiding wasted time and materials from painting a poor fit.

#353 · Repair Vehicle Bodies and Components

What should a technician do if a replacement panel's mounting holes are misaligned?

Answer

The technician should investigate whether the panel or the vehicle's mounting points need adjustment rather than forcing the parts together.

#354 · Repair Vehicle Bodies and Components

Why are both static and dynamic panel checks performed?

Answer

A panel can measure correctly at rest but still bind or rub during operation, so both checks are needed to confirm full function.

#355 · Repair Vehicle Bodies and Components

What is the risk of using a vacuum cup puller on deep structural damage?

Answer

A vacuum cup puller is meant for shallow, non-structural dents; it is not reliable for removing severe or structural damage.

#356 · Repair Vehicle Bodies and Components

Why is corrosion protection required even on small exposed areas like ground stud marks?

Answer

Any bare metal, regardless of size, is vulnerable to rust once exposed, so consistent corrosion protection is required everywhere it occurs.

#357 · Repair Vehicle Bodies and Components

What does a guide coat remaining after sanding indicate?

Answer

Guide coat still visible after sanding indicates a low area that needs more filler or sanding before the surface is truly flat.

#358 · Repair Vehicle Bodies and Components

Why is featheredging important to the final paint finish?

Answer

A poorly featheredged edge leaves a visible ridge that will show through new paint instead of blending into the original finish.

#359 · Repair Vehicle Bodies and Components

What is the purpose of block sanding with progressively finer grits?

Answer

Block sanding with finer grits removes high and low spots evenly and eliminates coarser scratch patterns before the panel is primed.

#360 · Repair Vehicle Bodies and Components

Why must sand scratches be checked before applying primer?

Answer

Sand scratches can telegraph through primer and become visible in the final paint if not fully removed beforehand.

#361 · Repair Vehicle Bodies and Components

What does equal diagonal measurements after a repair confirm?

Answer

Equal diagonal measurements after structural pulling confirm the structure has been returned closer to its original factory shape.

#362 · Repair Vehicle Bodies and Components

Why is repair versus replace partly a cost decision?

Answer

When repair time would cost more than a replacement part, replacement can be the more practical choice for non-structural panels.

#363 · Repair Vehicle Bodies and Components

What should be verified before reinstalling weatherstripping and trim clips?

Answer

Panel alignment, gap, and flush fit should be confirmed correct first, since trim can mask a poor fit if installed too early.

#364 · Repair Vehicle Bodies and Components

Why is hand-feel inspection used alongside visual inspection on a finished repair?

Answer

Some defects, like subtle waves or fine scratches, are easier to feel than see, so combining both methods catches more problems.

#365 · Repair Vehicle Bodies and Components

What is the significance of paint bubbling along a seam months after a repair?

Answer

Paint bubbling along a seam later usually signals incomplete or missing corrosion protection that allowed rust to form underneath.

#366 · Repair Vehicle Bodies and Components

Why does the technician check panel operation, not just static gaps?

Answer

A panel could measure correctly closed yet still rub or bind when opened, so functional operation must be checked too.

#367 · Repair Vehicle Bodies and Components

What is the purpose of remeasuring structure a second time after pulling?

Answer

The second measurement confirms the structural pull actually returned the vehicle to specification, not just that it looks straight by eye.

#368 · Repair Vehicle Bodies and Components

Why is documentation treated as a required verification step, not paperwork only?

Answer

Documentation gives a clear repair history useful if the vehicle is in a future collision, protecting both the shop and the customer.

#369 · Repair Vehicle Bodies and Components

What factor makes a structural panel repair decision different from a cosmetic panel decision?

Answer

A structural panel's role in the crash energy path and heat sensitivity limit repair options in ways a cosmetic panel does not face.

#370 · Repair Vehicle Bodies and Components

Why should thicker filler layers be avoided even to save time?

Answer

Thicker layers generate more curing heat and are more likely to crack later, so the durability cost outweighs any short term time savings.

#371 · Repair Plastics and Fibreglass

What does the resin code TPO stand for?

Answer

Thermoplastic olefin, a flexible plastic commonly used for bumper covers.

#372 · Repair Plastics and Fibreglass

What does the resin code ABS stand for?

Answer

Acrylonitrile butadiene styrene, a harder, more rigid plastic used in trim and some panels.

#373 · Repair Plastics and Fibreglass

What is a burn test used for in plastic repair?

Answer

Cutting a small scrap sliver and burning it to identify plastic type by flame color and smoke smell.

#374 · Repair Plastics and Fibreglass

What is a flex test used for?

Answer

A non-destructive hand-bend check on a hidden section to see if a plastic is flexible or rigid.

#375 · Repair Plastics and Fibreglass

What defines a thermoplastic?

Answer

A plastic that softens when heated and hardens again when cooled, repeatedly.

#376 · Repair Plastics and Fibreglass

What defines a thermoset?

Answer

A plastic that cures permanently and cannot be melted and reshaped again.

#377 · Repair Plastics and Fibreglass

Can thermoset plastics be plastic welded?

Answer

No, welding only works on thermoplastics; thermosets need adhesive bonding or layup.

#378 · Repair Plastics and Fibreglass

What are the two common plastic welding heat sources?

Answer

Hot air welders and airless welders with a heated tip.

#379 · Repair Plastics and Fibreglass

Why V-groove a crack before plastic welding?

Answer

It creates a channel for the filler rod and increases bonding surface area.

#380 · Repair Plastics and Fibreglass

What happens if you weld too fast?

Answer

The joint ends up weak and poorly fused.

#381 · Repair Plastics and Fibreglass

When is a welded backing strip used?

Answer

On cracks near mounting points that see repeated stress, for extra strength.

#382 · Repair Plastics and Fibreglass

How do two-part structural adhesives cure?

Answer

Through a chemical reaction between a resin component and a hardener.

#383 · Repair Plastics and Fibreglass

What is pot life?

Answer

The limited working time after mixing before an adhesive becomes too stiff to spread.

#384 · Repair Plastics and Fibreglass

What is the correct surface prep order before adhesive bonding?

Answer

Clean with plastic cleaner, abrade the surface, then apply adhesion promoter.

#385 · Repair Plastics and Fibreglass

Why apply an adhesion promoter to TPO before bonding?

Answer

It improves bonding on TPO's naturally low surface energy.

#386 · Repair Plastics and Fibreglass

What is the purpose of fibreglass mesh reinforcement?

Answer

To spread stress over a wider area instead of concentrating it at the crack line.

#387 · Repair Plastics and Fibreglass

What should be done after laying mesh into wet adhesive?

Answer

Apply a second layer of adhesive over the top to fully saturate the mesh.

#388 · Repair Plastics and Fibreglass

Why are trapped air bubbles under mesh a problem?

Answer

They weaken the repair and can lead to future failure.

#389 · Repair Plastics and Fibreglass

How is a large missing section of a bumper cover rebuilt?

Answer

With layered mesh and adhesive built up and shaped to match the original contour.

#390 · Repair Plastics and Fibreglass

What are fibreglass panels built from before resin is added?

Answer

Layers of glass fibre cloth or mat.

#391 · Repair Plastics and Fibreglass

What advantage does epoxy resin have over polyester resin?

Answer

Better bonding to cured fibreglass and less shrinkage while curing.

#392 · Repair Plastics and Fibreglass

What is the first step after removing damaged fibreglass material?

Answer

Grind a bevelled taper around the damaged area.

#393 · Repair Plastics and Fibreglass

Why are fibreglass cloth layers cut progressively larger?

Answer

To build the repair in a stepped shape matching the original panel thickness.

#394 · Repair Plastics and Fibreglass

Why roll or brush each resin-saturated cloth layer?

Answer

To remove trapped air bubbles that would otherwise create weak spots.

#395 · Repair Plastics and Fibreglass

What happens if you sand fibreglass resin before it fully cures?

Answer

The resin smears instead of cutting cleanly and releases more airborne fumes.

#396 · Repair Plastics and Fibreglass

What is the purpose of gelcoat?

Answer

To give a finished appearance and protect the fibreglass from moisture and UV.

#397 · Repair Plastics and Fibreglass

Why is gelcoat applied slightly thicker than the final thickness?

Answer

Because sanding and polishing afterward will remove some material.

#398 · Repair Plastics and Fibreglass

What comes after cured gelcoat before final buffing?

Answer

Wet sanding through progressively finer grits to blend the repair.

#399 · Repair Plastics and Fibreglass

Why must filler on flexible plastics stay flexible?

Answer

The panel bends during driving, and rigid material would crack loose.

#400 · Repair Plastics and Fibreglass

What happens if rigid filler is used on a flexible TPO bumper?

Answer

The filler eventually cracks and separates once the bumper flexes.

#401 · Repair Plastics and Fibreglass

What must be removed from a new flexible plastic part before bonding?

Answer

Mold release agent residue, using a plastic-specific pre-cleaner.

#402 · Repair Plastics and Fibreglass

What happens if pre-cleaning is skipped before bonding?

Answer

Adhesion failure results because release agent residue blocks proper grip.

#403 · Repair Plastics and Fibreglass

How is a flex test performed on a completed repair?

Answer

Gently flex the repaired area by hand, checking for cracking or separation.

#404 · Repair Plastics and Fibreglass

What does tap testing detect?

Answer

Voids or areas where resin or adhesive did not properly bond.

#405 · Repair Plastics and Fibreglass

What does a dull, hollow sound during tap testing mean?

Answer

There is likely a void where bonding did not occur properly.

#406 · Repair Plastics and Fibreglass

What should happen if a repair fails flex testing?

Answer

The repair should be reworked before it proceeds any further.

#407 · Repair Plastics and Fibreglass

Why keep plastic repair sanding discs separate from metal repair discs?

Answer

Metal filings trapped in plastic can telegraph through the finished paint.

#408 · Repair Plastics and Fibreglass

Why does a headlight lens use polycarbonate instead of TPO?

Answer

Polycarbonate offers the clarity and impact resistance a lens needs.

#409 · Repair Plastics and Fibreglass

What does the abbreviation SMC stand for?

Answer

Sheet molded compound, a reinforced thermoset panel material.

#410 · Repair Plastics and Fibreglass

What safety practice matters most during grinding, welding, and resin work?

Answer

Good ventilation combined with proper respiratory protection.

#411 · Repair Glass and Related Components

What is laminated glass made of?

Answer

Two layers of glass bonded to a soft plastic interlayer called polyvinyl butyral (PVB), which holds the glass together after an impact instead of letting it shatter apart.

#412 · Repair Glass and Related Components

Where is laminated glass used on a vehicle and why?

Answer

Windshields, because the driver needs a usable view after impact and the interlayer helps keep occupants inside the vehicle during a rollover.

#413 · Repair Glass and Related Components

What is tempered glass and how is it made?

Answer

A single sheet of glass heated and rapidly cooled, putting the outer surfaces under compression, which makes it about four to five times stronger than regular glass.

#414 · Repair Glass and Related Components

Where is tempered glass used and why?

Answer

Side windows and backlights, because it shatters into small rounded pieces instead of long sharp shards, reducing cutting risk.

#415 · Repair Glass and Related Components

Can tempered side glass ever be chip repaired with resin?

Answer

No. The internal stress pattern becomes unstable once the surface breaks, so any crack or chip in tempered glass requires full replacement.

#416 · Repair Glass and Related Components

Where can you find the manufacturer and glass type of a piece of automotive glass?

Answer

In the etched marking near the edge of the glass, which lists the manufacturer, DOT number, and whether the glass is laminated or tempered.

#417 · Repair Glass and Related Components

What is the pinch weld?

Answer

The flanged metal edge of the vehicle body opening that the glass and urethane adhesive bond to structurally.

#418 · Repair Glass and Related Components

What does a chalky white, cracked urethane bead usually indicate?

Answer

Age and UV damage to the existing adhesive, which can lead to leaks or a weakened bond.

#419 · Repair Glass and Related Components

What is the function of a rain sensor?

Answer

An optical module behind the rearview mirror that detects water on the glass and triggers the wipers automatically.

#420 · Repair Glass and Related Components

What is an ADAS camera bracket used for?

Answer

It holds a forward facing camera used for lane departure warning, automatic emergency braking, and adaptive cruise control.

#421 · Repair Glass and Related Components

What should you check first if only part of a backlight defroster grid stops heating?

Answer

The grid lines for a physical break in the glass, since this is often mistaken for an electrical fault.

#422 · Repair Glass and Related Components

What general chip size is a good candidate for resin injection repair?

Answer

Smaller than the size of a quarter, located away from the edge of the glass and outside the driver's primary sight line.

#423 · Repair Glass and Related Components

Why does a crack that reaches the edge of the windshield require replacement?

Answer

It compromises the edge zone where the glass gets its structural strength from the urethane bond to the body.

#424 · Repair Glass and Related Components

Why might a shop replace instead of repair a small chip in the driver's sight line?

Answer

Even a repairable sized chip can leave slight optical distortion after resin repair, which is a safety concern directly in front of the driver.

#425 · Repair Glass and Related Components

A windshield leaks only at highway speed, not when parked. What does this suggest?

Answer

Wind pulling water through a small gap in the seal rather than gravity pulling water through a larger opening.

#426 · Repair Glass and Related Components

What should be checked first if an ADAS camera throws a calibration fault after glass replacement?

Answer

Whether the camera bracket was reattached in exactly its original position before assuming an electrical problem.

#427 · Repair Glass and Related Components

How does a resin injection tool prepare a chip before injecting resin?

Answer

It clamps over the damaged area and creates a vacuum to pull air and moisture out of the crack first.

#428 · Repair Glass and Related Components

What is the final finishing step after curing resin in a chip repair?

Answer

Scrape the surface flush with the surrounding glass and polish it smooth.

#429 · Repair Glass and Related Components

What is the cold knife windshield removal method?

Answer

A hooked, sharp blade worked through the urethane bead from outside the vehicle along the pinch weld, often used when the glass will be replaced.

#430 · Repair Glass and Related Components

What is the wire out windshield removal method?

Answer

A thin, high strength wire pulled through the urethane bead with handles inside and outside, gentler on the glass and used when the glass must come out intact.

#431 · Repair Glass and Related Components

What risk comes with using a power cut out tool at the wrong blade angle?

Answer

It can cut into the pinch weld metal and damage the corrosion protection, creating a future rust point.

#432 · Repair Glass and Related Components

What must be done if bare metal is exposed on the pinch weld during removal?

Answer

Treat it with an approved primer before applying new urethane, or corrosion will start under the new bond.

#433 · Repair Glass and Related Components

Why is old cured urethane trimmed to a thin layer instead of removed completely?

Answer

New urethane bonds better to a thin layer of cured urethane than to bare painted metal.

#434 · Repair Glass and Related Components

What preparation is required on the ceramic frit band before bonding new glass?

Answer

Clean it with the adhesive manufacturer's approved glass cleaner, then apply an approved primer.

#435 · Repair Glass and Related Components

What shape should a properly applied urethane bead form?

Answer

A continuous, consistent triangular bead following the height and width the adhesive manufacturer specifies.

#436 · Repair Glass and Related Components

Is safe drive away time the same for every urethane product?

Answer

No. It changes from product to product and shifts with temperature and humidity, so the specific manufacturer's stated time must always be followed.

#437 · Repair Glass and Related Components

Why does urethane cure time change with humidity?

Answer

Urethane cures through a chemical reaction with moisture in the air, so more or less humidity changes the cure speed.

#438 · Repair Glass and Related Components

What is the main risk of releasing a vehicle before its safe drive away time has passed?

Answer

The windshield may not yet be able to do its structural job in a collision, since the bond has not fully developed strength.

#439 · Repair Glass and Related Components

Why should new clips always be used when reinstalling moulding?

Answer

Old clips become brittle with age and heat cycling, and a broken clip can let moulding lift at highway speed.

#440 · Repair Glass and Related Components

When should new moulding be fitted into the urethane bead?

Answer

Before the urethane fully skins over, so it seats correctly into the wet bead and the reveal line stays even.

#441 · Repair Glass and Related Components

What should be replaced when reinstalling a rain sensor after glass replacement?

Answer

The optical gel pad that makes direct contact with the inside surface of the glass.

#442 · Repair Glass and Related Components

Why does an ADAS camera need recalibration after windshield replacement?

Answer

Its calibration data assumes a specific relationship between the lens and the road, and new glass can shift that relationship.

#443 · Repair Glass and Related Components

What is static ADAS calibration?

Answer

Using printed target boards set at precise measured distances and heights in a level, controlled shop environment along with a scan tool.

#444 · Repair Glass and Related Components

What is dynamic ADAS calibration?

Answer

Driving the vehicle on the road at a specified speed for a set distance while a scan tool monitors the system.

#445 · Repair Glass and Related Components

Why should a technician confirm the exact calibration method for each vehicle instead of assuming?

Answer

Some vehicles require both static and dynamic calibration together, not just one method.

#446 · Repair Glass and Related Components

What documentation should be kept after a successful ADAS recalibration?

Answer

A scan tool printout confirming the calibration was completed successfully, kept with the job file.

#447 · Repair Glass and Related Components

What is the standard method for checking a new windshield for leaks?

Answer

Running water over the top and sides of the glass with a hose while an assistant inside checks for water entry.

#448 · Repair Glass and Related Components

Why wait a minute or two at each section during a water leak test?

Answer

Urethane leaks are often slow seeps rather than instant streams, so rushing the test can miss a leak.

#449 · Repair Glass and Related Components

What alternative test can find small leak paths that a water test might miss?

Answer

A smoke or air pressure test inside the closed cabin, particularly useful around corners where the bead is hardest to apply evenly.

#450 · Repair Glass and Related Components

What should be checked before releasing a vehicle after any glass replacement involving a rain sensor or ADAS camera?

Answer

Run the wipers with water to confirm the rain sensor works, and confirm the scan tool shows a completed ADAS calibration.

#451 · Refinish Vehicle Bodies Trim and Components

What is a single-stage paint finish?

Answer

A paint that carries both colour and gloss in one product, sprayed and cured as a single layer without a separate clearcoat.

#452 · Refinish Vehicle Bodies Trim and Components

What is a basecoat clearcoat system?

Answer

A two-layer system where the basecoat carries colour only and a separate clearcoat is sprayed over top to provide gloss and UV protection.

#453 · Refinish Vehicle Bodies Trim and Components

Why does basecoat dry with no gloss?

Answer

The basecoat only carries pigment and metallic or pearl effect; gloss and depth come entirely from the clearcoat sprayed afterward.

#454 · Refinish Vehicle Bodies Trim and Components

What is a tri-coat paint system?

Answer

A three-layer system that adds a midcoat, or pearl coat, containing mica particles between the basecoat and clearcoat.

#455 · Refinish Vehicle Bodies Trim and Components

Why are tri-coat pearl finishes harder to match?

Answer

The midcoat must be sprayed with very even coats; too many passes darken the shade and too few leave it looking pale.

#456 · Refinish Vehicle Bodies Trim and Components

What is a spectrophotometer used for in refinishing?

Answer

An electronic tool that reads how existing paint reflects light to suggest the closest matching colour formula.

#457 · Refinish Vehicle Bodies Trim and Components

What is a colour chip used for?

Answer

A small painted sample on a fan deck held next to the vehicle for a fast first visual check of a colour formula.

#458 · Refinish Vehicle Bodies Trim and Components

What is a test panel spray-out?

Answer

A small batch of matched paint sprayed on a scrap panel with the planned gun setup, then compared to the vehicle in daylight.

#459 · Refinish Vehicle Bodies Trim and Components

Why check metallic colour from more than one angle?

Answer

Metal flake reflects light differently depending on angle, so a match can look correct straight on but off from the side.

#460 · Refinish Vehicle Bodies Trim and Components

What does blending accomplish in a repair?

Answer

It feathers new paint into surrounding original paint so there is no visible line where the repair ends.

#461 · Refinish Vehicle Bodies Trim and Components

How far should clearcoat extend compared to basecoat during a blend?

Answer

The clearcoat is blended further out than the basecoat, usually covering the entire adjacent panel.

#462 · Refinish Vehicle Bodies Trim and Components

What is a blending solvent used for?

Answer

It melts the outer edge of new paint into the old paint so the transition is smooth and invisible.

#463 · Refinish Vehicle Bodies Trim and Components

Where should a technician plan to break a blend line?

Answer

At a natural body line, trim piece, or panel gap on the vehicle to help hide the transition.

#464 · Refinish Vehicle Bodies Trim and Components

What is mil buildup?

Answer

The thickness of the applied paint film, measured in thousandths of an inch with a mil gauge.

#465 · Refinish Vehicle Bodies Trim and Components

What spray gun distance is generally recommended?

Answer

Around six to eight inches, held at a consistent distance throughout each pass.

#466 · Refinish Vehicle Bodies Trim and Components

What overlap percentage keeps paint film even between passes?

Answer

Roughly fifty percent overlap so every area receives paint from two overlapping passes.

#467 · Refinish Vehicle Bodies Trim and Components

What is flash time?

Answer

The waiting period between coats that allows solvents to evaporate before the next coat is applied.

#468 · Refinish Vehicle Bodies Trim and Components

What happens if the next coat is sprayed before enough flash time passes?

Answer

Solvent gets trapped underneath, which can cause the paint to wrinkle or blister.

#469 · Refinish Vehicle Bodies Trim and Components

What is orange peel?

Answer

A bumpy surface texture resembling citrus skin, usually caused by paint drying too fast before it can level out.

#470 · Refinish Vehicle Bodies Trim and Components

What causes runs and sags?

Answer

Too much paint applied in one spot, often from moving the gun too slowly or holding it too close, without enough flash time.

#471 · Refinish Vehicle Bodies Trim and Components

What is fisheye and what causes it?

Answer

A small circular crater in the paint, most often caused by silicone contamination from wax, polish, or tire dressing.

#472 · Refinish Vehicle Bodies Trim and Components

What is the real fix for fisheye contamination?

Answer

Properly cleaning and degreasing the surface before spraying, not just adding fisheye eliminator additive each time.

#473 · Refinish Vehicle Bodies Trim and Components

What is blush and what causes it?

Answer

A hazy, milky cloudiness in the paint film caused by moisture in the compressed air or overly humid spraying conditions.

#474 · Refinish Vehicle Bodies Trim and Components

What prevents blush from occurring?

Answer

A working air dryer and moisture trap fitted on the compressed air supply system.

#475 · Refinish Vehicle Bodies Trim and Components

What is solvent pop?

Answer

Small pinholes or bubbles caused by trapped solvent trying to escape a coat that skinned over too fast.

#476 · Refinish Vehicle Bodies Trim and Components

What are dirt nibs?

Answer

Small bumps in the paint caused by dust or debris landing in the wet film during spraying.

#477 · Refinish Vehicle Bodies Trim and Components

Why must clearcoat be fully cured before buffing?

Answer

A fully cured clearcoat resists scratching during buffing far better than a clearcoat that only feels dry to the touch.

#478 · Refinish Vehicle Bodies Trim and Components

What speeds up clearcoat curing in a shop?

Answer

A baking booth or infrared lamps are commonly used to speed the curing process.

#479 · Refinish Vehicle Bodies Trim and Components

What is buffing used for on a repaired panel?

Answer

Removing very fine surface defects such as light orange peel or dust nibs without cutting through the colour coat underneath.

#480 · Refinish Vehicle Bodies Trim and Components

What is buffer burn?

Answer

A dull or hazy patch caused by removing too much clearcoat in one spot during aggressive buffing.

#481 · Refinish Vehicle Bodies Trim and Components

What technique helps avoid buffer burn?

Answer

Working in overlapping passes and checking the surface often rather than staying in one spot too long.

#482 · Refinish Vehicle Bodies Trim and Components

What comes after compounding to remove haze?

Answer

A finer polish is used to remove the light haze left by the compound and restore full gloss and clarity.

#483 · Refinish Vehicle Bodies Trim and Components

Why wait before waxing a freshly buffed clearcoat?

Answer

Many manufacturers recommend waiting so the clearcoat can finish off-gassing remaining solvents before wax is applied.

#484 · Refinish Vehicle Bodies Trim and Components

Why is natural daylight preferred for a final colour check?

Answer

Fluorescent shop lighting can make two different colours look identical, hiding a real mismatch that daylight reveals.

#485 · Refinish Vehicle Bodies Trim and Components

From how many angles should a repaired panel be inspected?

Answer

Straight on, from the side, and from above if possible, since metallic colours can look correct from only one angle.

#486 · Refinish Vehicle Bodies Trim and Components

What is a cross light used for?

Answer

A raking light held at a low angle to reveal surface texture problems invisible under normal overhead lighting.

#487 · Refinish Vehicle Bodies Trim and Components

How is gloss level checked between a repaired panel and its neighbours?

Answer

By comparing reflection sharpness; a dull or hazy patch signals undercured clearcoat or a buffing mistake.

#488 · Refinish Vehicle Bodies Trim and Components

What should be checked about trim and hardware before releasing a vehicle?

Answer

That all trim pieces, emblems, and mirrors removed for the repair have been reinstalled correctly.

#489 · Refinish Vehicle Bodies Trim and Components

Why is an adhesion promoter required on bare plastic trim?

Answer

Paint does not stick well to bare plastic without an adhesion promoter applied first.

#490 · Refinish Vehicle Bodies Trim and Components

What is a common cause of paint peeling off plastic trim months later?

Answer

Skipping the adhesion promoter step before applying basecoat to bare plastic.

#491 · Refinish Vehicle Bodies Trim and Components

How should a small removed trim part be refinished?

Answer

Mounted on a stand or hook so the technician can rotate it for even coverage on all sides.

#492 · Refinish Vehicle Bodies Trim and Components

What additive helps paint flex with plastic trim?

Answer

A flexible additive mixed into the paint so the coating can bend with the plastic without cracking.

#493 · Refinish Vehicle Bodies Trim and Components

What should be recorded on the repair order after a refinish job?

Answer

The paint code used, any custom tint adjustment, and how far the blend extended.

#494 · Refinish Vehicle Bodies Trim and Components

Why are photographs taken of a completed refinish repair?

Answer

They support the shop's quality records and any insurance or warranty documentation needed later.

#495 · Refinish Vehicle Bodies Trim and Components

What effect does a clogged spray booth intake filter have?

Answer

It reduces airflow and can pull unfiltered dust in around the booth door seals, contaminating panels.

#496 · Refinish Vehicle Bodies Trim and Components

What step removes fine dust right before spraying basecoat?

Answer

Wiping the panel with a tack cloth immediately after the final compressed air blow down.

#497 · Refinish Vehicle Bodies Trim and Components

Why does a worn paint suit cause contamination?

Answer

It can shed loose fibres and lint directly onto a freshly sprayed panel inside the booth.

#498 · Refinish Vehicle Bodies Trim and Components

Where is the vehicle paint code sticker usually located?

Answer

Inside the door jamb, trunk area, or engine bay of the vehicle.

#499 · Refinish Vehicle Bodies Trim and Components

What effect does a low spray booth temperature have on clearcoat?

Answer

It slows curing and can allow solvent to become trapped underneath the paint surface.

#500 · Refinish Vehicle Bodies Trim and Components

Where should a final overspray and masking residue check focus?

Answer

Glass, trim pieces, and adjacent panels located near the repaired area of the vehicle.